

SED & Vi/Vim

Editor Commands

Complete Reference Guide for Linux / Unix OS

Covers: sed stream editor • Vi normal/insert/command modes • Vim extensions

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PART 1 — SED (Stream Editor)

1. Introduction & Basic Syntax

sed (stream editor) reads input line-by-line, applies editing commands, and writes to stdout. Original files are unchanged unless -i is used.

Command	Description
sed [options] 'cmd' file	Apply cmd to file
sed -n 'cmd' file	Suppress auto-print (silent mode)
sed -i 'cmd' file	Edit file in-place
sed -i.bak 'cmd' file	In-place edit with .bak backup
sed -e 'cmd1' -e 'cmd2' file	Multiple commands
sed -f script.sed file	Read commands from script file
sed -r 'cmd' file	Use extended regex (ERE) — GNU sed
sed -E 'cmd' file	Use extended regex (POSIX compatible)

2. Address Ranges

Addresses restrict which lines a command acts on.

Command	Description
sed '3 cmd' file	Apply to line 3 only
sed '3,7 cmd' file	Apply to lines 3–7
sed '3,\$ cmd' file	Apply from line 3 to end
sed '1~2 cmd' file	Apply to every 2nd line starting at 1 (GNU)
sed '/pattern/ cmd' file	Apply to lines matching pattern
sed '/start/,/end/ cmd' file	Apply between matching lines
sed '3,/end/ cmd' file	From line 3 to next match of /end/
sed '/pat!/ cmd' file	Apply to lines NOT matching pattern
sed '\$d' file	Address \$ = last line

3. Substitution Command (s)

Command	Description
s/old/new/	Replace first occurrence per line
s/old/new/g	Replace ALL occurrences per line
s/old/new/2	Replace 2nd occurrence per line
s/old/new/ig	Case-insensitive, replace all
s/old/new/p	Print line if substitution made

Command	Description
s/old/new/w out.txt	Write changed lines to out.txt
s/^/PREFIX/	Prepend text to every line
s/\$/ SUFFIX/	Append text to every line
s/.*/(match)/	Replace entire line
s/\\(group\\)/[\\1]/	Back-reference with \\1 (BRE)
s/(group)/[\\1]/	Back-reference with \\1 (ERE, -E)
s/&/replaced/	& refers to entire matched string
s /old/path /new/path g	Use as delimiter (avoids escaping /)

4. Deletion & Insertion Commands

Command	Description
d	Delete current line
D	Delete first line of multi-line pattern space
/pattern/d	Delete lines matching pattern
i\\text	Insert text BEFORE current line
a\\text	Append text AFTER current line
c\\text	Replace (change) current line with text
2i\\text	Insert text before line 2
\$a\\text	Append text after last line

5. Print, Quit & File Commands

Command	Description
p	Print current line (use with -n to print only matches)
P	Print first line of multi-line pattern space
=	Print current line number
q	Quit after current line
Q	Quit without printing (GNU sed)
r filename	Read and append file after current line
R filename	Read one line from file (GNU sed)
w filename	Write pattern space to file
W filename	Write first line of pattern space to file
l	Print line unambiguously (show non-printables)

6. Labels, Branching & Hold Space

Command	Description
:label	Define a label
b label	Branch (jump) to label
b	Branch to end of script (skip remaining)
t label	Branch if substitution made since last t
T label	Branch if NO substitution made (GNU sed)
n	Read next line into pattern space
N	Append next line to pattern space
h	Copy pattern space to hold space
H	Append pattern space to hold space
g	Copy hold space to pattern space
G	Append hold space to pattern space
x	Exchange pattern and hold spaces
y/abc/xyz/	Transliterate chars (a→x, b→y, c→z)

7. Practical sed Examples

Command	Description
sed -n '5,10p' file	Print lines 5 to 10
sed '/^#/d' file	Remove comment lines
sed '/^\$/d' file	Remove blank lines
sed 's/ */ /g' file	Squeeze multiple spaces into one
sed 's/^[\t]*//'	Remove leading whitespace
sed 's/[\t]*\$//'	Remove trailing whitespace
sed -n '/ERROR/p' log	Print lines containing ERROR
sed '1d' file	Remove first line (header)
sed -i 's/foo/bar/g' *.txt	In-place replace in all .txt files
sed -n '\$p' file	Print last line
sed 'G' file	Double-space a file
sed '=' file sed 'N;s/\n/\t/'	Add line numbers
sed '/START/,/END/d' file	Delete block between patterns
sed -n '/pat/,/end/p' file	Print block between patterns

PART 2 — Vi / Vim Editor

8. Vi/Vim Modes Overview

Command / Key	Description
Normal (Command) Mode	Default mode. Navigate and run commands. Press Esc to return here.
Insert Mode	Type/edit text. Enter via i, I, a, A, o, O, s, S, R.
Visual Mode	Select text visually. v=char, V=line, Ctrl+v=block.
Command-Line (Ex) Mode	Enter : from Normal mode. Run Ex commands, search, substitute.
Replace Mode	Overwrite characters. Enter via R.

9. Opening & Saving Files

Command / Key	Description
vi file.txt	Open file.txt in Vi
vim file.txt	Open file.txt in Vim
vi +n file	Open file at line n
vi +/pattern file	Open file at first match of pattern
vi -R file	Open in read-only mode
:w	Save (write) current file
:w file.txt	Save to file.txt
:wq or ZZ	Save and quit
:q	Quit (fails if unsaved changes)
:q!	Force quit without saving
:wq!	Force save and quit (read-only override)
:x	Save only if changed, then quit
:e file	Open another file in current window
:e!	Re-load current file, discard changes
:saveas new.txt	Save as new filename (Vim)

10. Navigation — Normal Mode

Command / Key	Description
h / j / k / l	← ↓ ↑ → (character movement)
w / W	Next word start (w=word, W=WORD incl. punctuation)
b / B	Previous word start
e / E	End of current word
0 (zero)	Beginning of line

Command / Key	Description
^	First non-blank character of line
\$	End of line
gg	Go to first line of file
G	Go to last line
nG or :n	Go to line n
Ctrl+f / Ctrl+b	Page forward / Page backward
Ctrl+d / Ctrl+u	Half-page down / up
Ctrl+e / Ctrl+y	Scroll down / up one line
H / M / L	Top / Middle / Bottom of visible screen
{ / }	Previous / Next paragraph
(/)	Previous / Next sentence
%	Jump to matching bracket / parenthesis
fx / Fx	Find char x forward / backward on line
tx / Tx	Move to just before char x (forward/backward)
; / ,	Repeat last f/t forward / backward
* / #	Search word under cursor forward / backward
zz / zt / zb	Redraw: cursor at center / top / bottom

11. Editing Commands

Command / Key	Description
i / I	Insert before cursor / at line start
a / A	Append after cursor / at line end
o / O	Open new line below / above and enter Insert
s / S	Substitute char / entire line
r / R	Replace one char / enter Replace mode
x / X	Delete char at / before cursor
dd	Delete (cut) current line
ndd	Delete n lines
D	Delete from cursor to end of line
dw / db / de	Delete word forward / backward / to end
d0 / d\$	Delete to start / end of line
dgg / dG	Delete to file start / end
cc / C	Change entire line / to end of line
cw / cb	Change word forward / backward

Command / Key	Description
c0 / c\$	Change to start / end of line
J	Join current line with next (remove newline)
gJ	Join without inserting space
~	Toggle case of character under cursor
gU{motion}	Make motion text UPPERCASE (Vim)
gu{motion}	Make motion text lowercase (Vim)
>> / <<	Indent / Un-indent current line
n>>	Indent n lines
=	Auto-indent (Vim)
.	Repeat last change

12. Search & Replace

Command / Key	Description
/pattern	Search forward for pattern
?pattern	Search backward for pattern
n / N	Next / Previous match
:%s/old/new/g	Replace all occurrences in entire file
:%s/old/new/gc	Replace all with confirmation
:s/old/new/	Replace first on current line
:s/old/new/g	Replace all on current line
:'<, '>s/old/new/g	Replace in visual selection
:5,10s/old/new/g	Replace in lines 5–10
:%s/\bword\b/new/g	Whole-word match replace
:%s/old/new/gi	Case-insensitive replace
:noh	Clear search highlight
:set hlsearch	Enable search highlight (Vim)
:set ignorecase	Case-insensitive search
:set smartcase	Case-sensitive only if uppercase used

13. Copy, Cut & Paste

Command / Key	Description
yy or Y	Yank (copy) current line
nyy	Yank n lines
yw / yb	Yank word forward / backward
y\$ / y0	Yank to end / start of line
yG / yg	Yank to end / start of file
p / P	Paste after / before cursor
dd	Cut current line (also deletes)
dw	Cut word
"ayy	Yank line into register a
"ap	Paste from register a
"+yy	Yank to system clipboard (Vim)
"+p	Paste from system clipboard
:reg	Show all registers

14. Undo, Redo & Marks

Command / Key	Description
u	Undo last change
U	Undo all changes on current line
Ctrl+r	Redo (Vim only)
nu	Undo n changes
ma	Set mark 'a' at cursor
`a	Jump to exact position of mark a
'a	Jump to start of line of mark a
``	Return to position before last jump
''	Return to line before last jump
:marks	List all marks
Ctrl+o / Ctrl+i	Jump older / newer position in jump list
:jumps	Show jump list

15. Windows & Buffer Commands (Vim)

Command / Key	Description
:sp file	Horizontal split, open file
:vsp file	Vertical split, open file
Ctrl+w s / v	Split horizontally / vertically
Ctrl+w h/j/k/l	Move to left/down/up/right window
Ctrl+w w	Cycle to next window
Ctrl+w q	Close current window
Ctrl+w =	Equalize window sizes
Ctrl+w + / -	Increase / Decrease window height
:ls or :buffers	List open buffers
:bn / :bp	Next / Previous buffer
:bd	Delete (close) current buffer
:b n	Switch to buffer n
:tabnew file	Open file in new tab
gt / gT	Next / Previous tab
:tabclose	Close current tab

16. Visual Mode

Command / Key	Description
v	Start character-wise visual selection

Command / Key	Description
V	Start line-wise visual selection
Ctrl+v	Start block (column) visual selection
o	Move to other end of selection
gv	Re-select last visual selection
d / x	Delete selected text
y	Yank (copy) selected text
c	Change (replace) selected text
> / <	Indent / Un-indent selection
~	Toggle case of selection
u / U	Make selection lowercase / UPPERCASE
:	Enter Ex command on selection (:'<,>)
I (block mode)	Insert text at start of each selected line
A (block mode)	Append text at end of each selected line

17. Ex / Colon Commands

Command / Key	Description
:n	Go to line n
:\$	Go to last line
:1,\$d	Delete all lines (clear file)
:r file	Read and insert file below cursor
:r !cmd	Insert output of shell command
:!cmd	Run shell command, return to Vi
:shell	Open shell (type exit to return)
:set nu	Show line numbers
:set nonu	Hide line numbers
:set list	Show invisible chars (tabs, EOL)
:set nolist	Hide invisible chars
:set wrap / nowrap	Enable / Disable line wrapping
:set tabstop=4	Set tab width to 4
:set expandtab	Use spaces instead of tabs
:set autoindent	Enable auto-indenting
:set syntax=python	Set syntax highlighting
:syntax on / off	Toggle syntax highlighting
:colorscheme name	Change color scheme
:help keyword	Open Vim help for keyword
:version	Show Vim version and compile options
:map key cmd	Map key to cmd in Normal mode
:imap key cmd	Map key in Insert mode
:ab abbr expansion	Create abbreviation
:cd /path	Change working directory
:pwd	Show current directory
:g/pattern/cmd	Run cmd on all matching lines
:g!/pattern/cmd	Run cmd on NON-matching lines
:g/pattern/d	Delete all lines matching pattern
:v/pattern/p	Print lines NOT matching pattern

18. Vim — Advanced Features

Command / Key	Description
Ctrl+n / Ctrl+p	Auto-complete word (Insert mode)

Command / Key	Description
Ctrl+x Ctrl+l	Line-completion (Insert mode)
Ctrl+x Ctrl+f	Filename completion (Insert mode)
qa	Start recording macro into register a
q	Stop recording macro
@a	Play macro in register a
@@	Repeat last macro
10@a	Play macro 10 times
:source file.vim	Execute Vim script file
:mksession sess.vim	Save session to file
vim -S sess.vim	Restore session
gd / gD	Go to local / global definition
K	Look up keyword under cursor (man page)
ga	Show ASCII value of character
g8	Show hex UTF-8 of character
Ctrl+g	Show file info and cursor position
:set spell	Enable spell check
]s / [s	Next / Previous misspelled word
z=	Suggest spelling corrections
:diffthis	Mark window for diff
:diffsplit file	Open file in diff split
do / dp	Diff obtain / Diff put (in diff mode)
:TOhtml	Convert buffer to HTML

Note: Commands prefixed with : are entered in Command-Line mode. n in commands means a number (e.g., 3dd deletes 3 lines). Ctrl+key means hold Control and press key. Vim-only features are noted where applicable; basic Vi may not support them.